

Multi-Source Candidate Data Transformer — Technical Design

Pipeline Workflow



Sources

- Structured:** Recruiter CSV (name, email, phone, current_company, title).
- Unstructured:** Resume PDF (pdfplumber text extraction + regex/section heuristics).

Canonical Schema

candidate_id, full_name, emails[], phones[] (E.164), location {city, region, country}, links {linkedin, github, portfolio, other}, headline, years_experience, skills[{name, confidence, sources}], experience[], education[], provenance[{field, source, method}], overall_confidence

Normalization

- Phones** → E.164 via *phonenumbers* library, default region India (IN).
- Dates** → YYYY-MM via *python-dateutil*.
- Skills** → canonical names via curated lookup dictionary (deterministic, explainable keyword mapping).

Merge & Confidence Logic

- Dedup Key:** Case-insensitive email overlap (names alone are too unreliable for unique candidate mapping).
- Confidence Scoring:** CSV (recruiter-entered) confidence: 0.9. Resume (heuristic extraction) confidence: 0.6.
- Agreement Boost:** Fields where both sources agree have their confidence boosted to 0.95.
- Conflict Resolution:** Higher-confidence source value (CSV) is kept, confidence drops to 0.5, and both values are logged in provenance metadata.
- Missing Resume Handling:** A missing or empty resume never crashes the pipeline or duplicates candidate profiles; it degrades gracefully, falling back to CSV-only records.

Runtime Config

The internal canonical record is always computed in full. A separate projector layer reshapes the output according to a JSON configuration (loaded from *input/config.json*), supporting: field subset selection; rename/remap via a 'from' path (dot and bracket notation, including array-map patterns like *skills[].name*); include/exclude toggles for provenance and confidence; and an *on_missing* policy (null / omit / error). If the configuration is missing or invalid, a warning is logged and the pipeline falls back to default settings without crashing.

Robustness & Validation

Every parser and file loader is wrapped in try/except blocks; missing files generate warnings and return empty datasets, never crashing. The validator checks schema constraints, verifies email format, and flags invalid phone formats before any output is exported, avoiding silent failures.

Assumptions & Descoped Items

- Default Region:** Default phone country code region is assumed to be India (IN).
- Experience & Education Structuring:** Deep structural parsing of experience and education fields was **intentionally descoped** due to the assignment's time budget. They are kept as raw lists of summaries with source metadata.
- LinkedIn & GitHub Integration:** Active profile scraping and third-party API integration were descoped. Instead, simple regex patterns extract social profile URLs from the resume text and map them under links.
- Resume Parsing:** Resume parsing is heuristic-based (regex + section detection), not ML-based. It performs well on standard layouts and may under-extract on unusual formats.